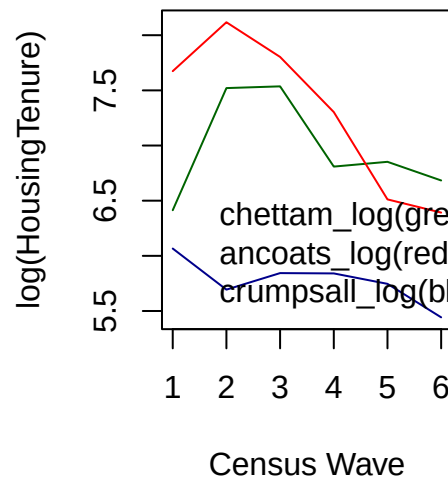
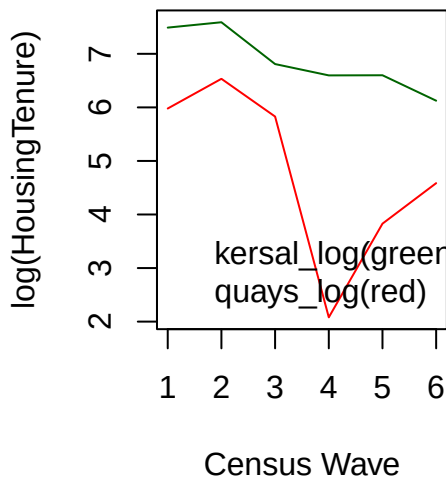


Someplots

```
library(readr)
library(janitor)
library(tidyverse)
library(patchwork)
library(modelsummary)
# Load HousingPanelwide data
HousingPanelwide <- read_csv("d:/PositStuff/gmpanel/HousingPanelwide.csv")
par(mfrow=c(1,2))
ts.plot(cbind(log(HousingPanelwide$kersal.Local Authority etc`),log(HousingPanelwide$`quays
  legend("bottomleft",legend = c("kersal_log(green)","quays_log(red)"),col=c("darkgreen","red")

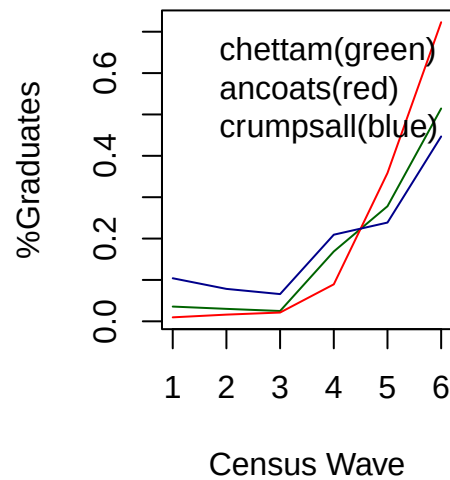
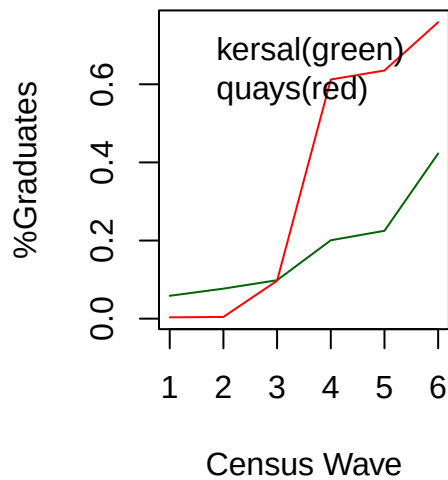
ts.plot(cbind(log(HousingPanelwide$`chettam.Local Authority etc`),log(HousingPanelwide$`anco
legend("bottomleft",legend = c("chettam_log(green)","ancoats_log(red)","crumpsall_log(blue)"))
```

Authority owned tenure 1971–2



```
EducationPanelwide <- read_csv("d:/PositStuff/gmpanel/EducationPanelwide.csv")
par(mfrow=c(1,2))
ts.plot(cbind(EducationPanelwide$`kersal.Graduate`/(EducationPanelwide$kersal.Graduate+EducationPanelwide$`quays.Graduate`),
             legend("topleft",legend = c("kersal(green)","quays(red)"),col=c("darkgreen","red"),bty = "n"))
ts.plot(cbind(EducationPanelwide$`chettam.Graduate`/(EducationPanelwide$chettam.Graduate+EducationPanelwide$`ancoats.Graduate`+EducationPanelwide$`crumpsall.Graduate`),
             legend("topleft",legend = c("chettam(green)","ancoats(red)","crumpsall(blue)"),col=c("darkgreen","red","blue"),bty = "n"))
```

Graduate education 1971–2021 Graduate education 1971–2021 c



```
EducationPanellong <- EducationPanelwide |>
# Create a row identifier
mutate(row_id = row_number()) |>

pivot_longer(
  cols = -c(1, row_id),
  names_to = c("location", "variable"),
  names_pattern = "^(.*?)\\.(.*)$" #No i do not know how to use pattern matching properly,
) |>

# Pivot wider while respecting the row_id and location mapping
pivot_wider(
  id_cols = c(row_id, location),
  names_from = variable,
  values_from = value
) |>

# 4. Clean up the temporary row_id column
select(-row_id) |>
janitor::clean_names()
```

```
HousingPanellong <- HousingPanelwide |>
# 1. Create a row identifier so each year stays separate
mutate(row_id = row_number()) |>

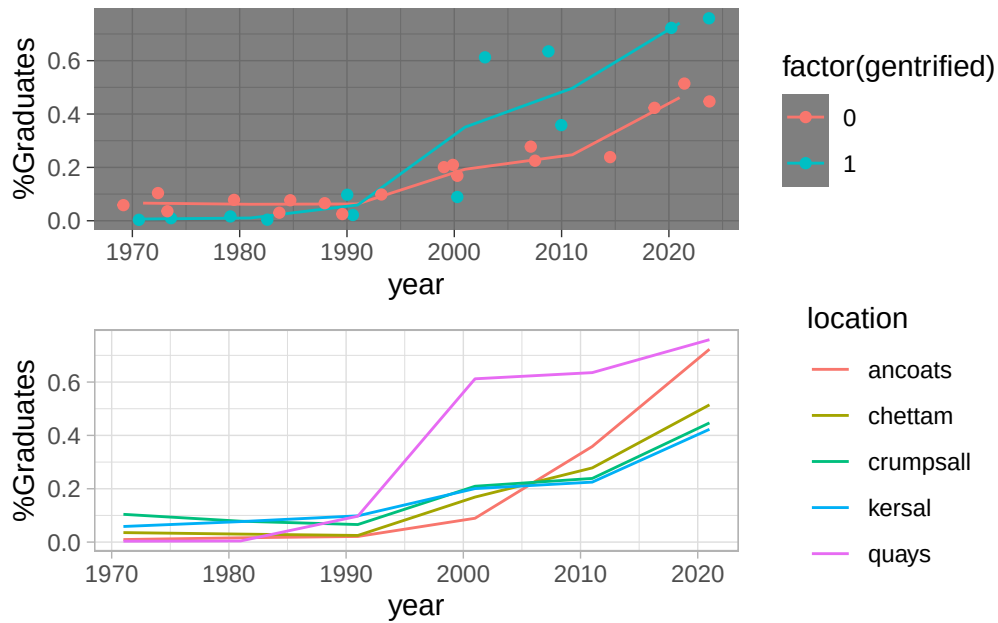
pivot_longer(
  cols = -c(1, row_id),
  names_to = c("location", "variable"),
  names_pattern = "^(.*)\\.\\.(.*)$"
) |>

pivot_wider(
  id_cols = c(row_id, location),
  names_from = variable,
  values_from = value
) |>

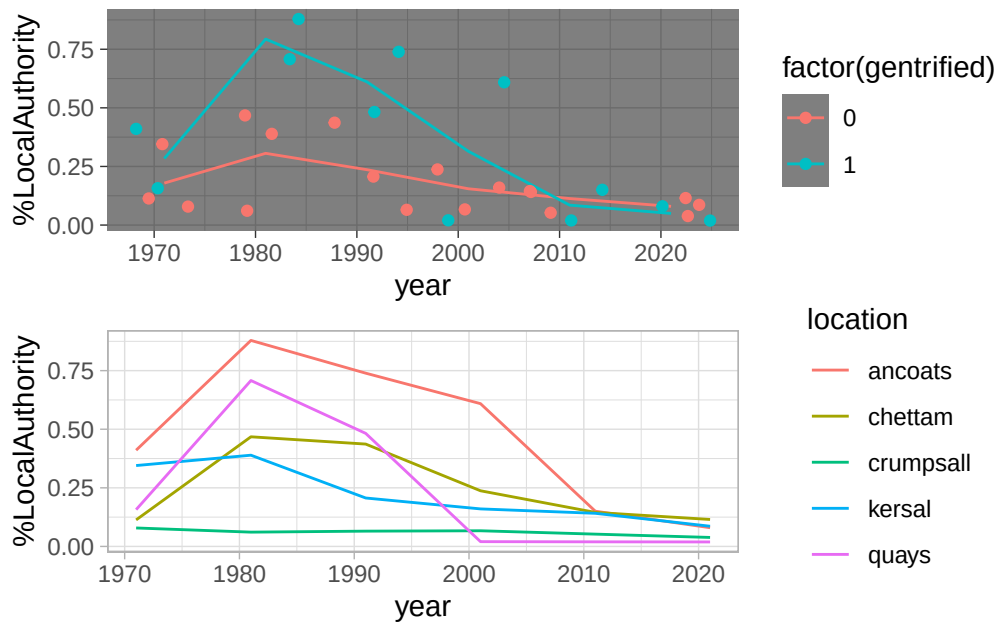
select(-row_id) |>
janitor::clean_names()
```

```
Panellong = merge(HousingPanellong, EducationPanellong) |>
mutate(gentrified=ifelse(location%in%c("ancoats", "quays"),1,0))
```

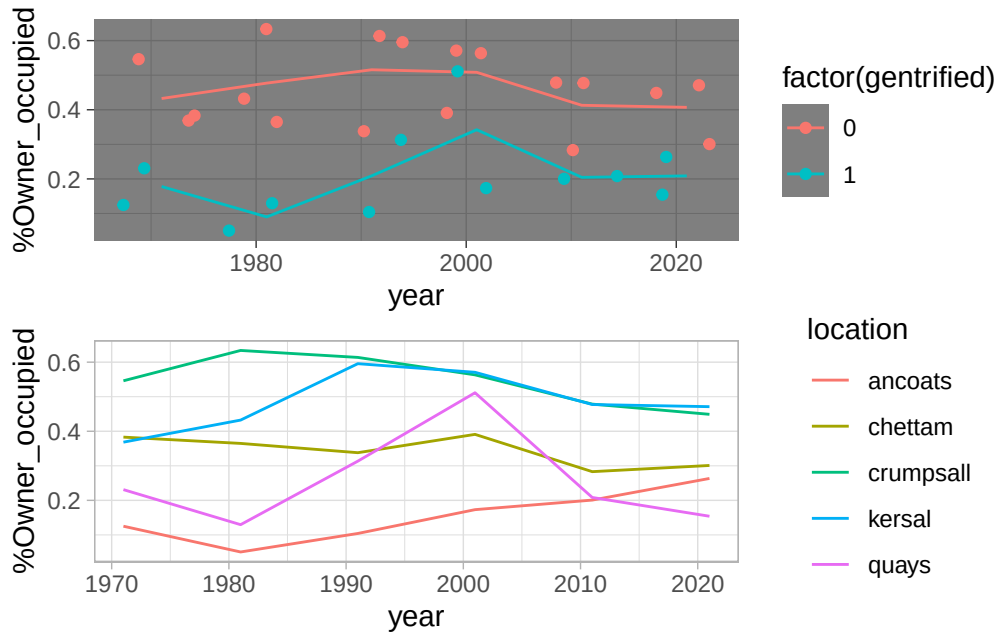
```
p1=ggplot(Panellong,aes(x=year,y=graduate/(graduate+not_graduate),colour=location))+geom_line()
p2=ggplot(Panellong,aes(x=year,y=graduate/(graduate+not_graduate),colour=factor(gentrified)))+geom_line()
p2/p1
```



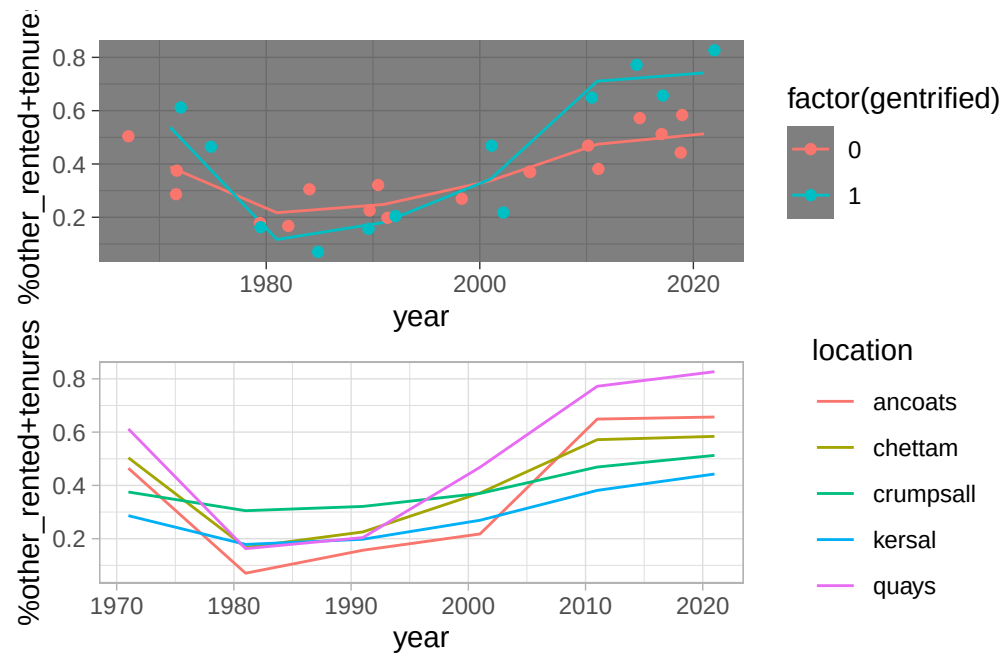
```
p3=ggplot(PanelLong,aes(x=year,y=local_authority_etc/(owner_occupied+other_rented+other_tenu
p4=ggplot(PanelLong,aes(x=year,y=local_authority_etc/(owner_occupied+other_rented+other_tenu
p4/p3
```



```
p5=ggplot(PanelLong,aes(x=year,y=owner_occupied/(owner_occupied+other_rented+other_tenures+1))
p6=ggplot(PanelLong,aes(x=year,y=owner_occupied/(owner_occupied+other_rented+other_tenures+1))
p6/p5
```



```
p7=ggplot(PanelLong,aes(x=year,y=(other_rented+other_tenures)/(owner_occupied+other_rented+other_tenures+1))
  ylab("%other_rented+tenures")+theme_light()
p8=ggplot(PanelLong,aes(x=year,y=(other_rented+other_tenures)/(owner_occupied+other_rented+other_tenures+1))
  stat_summary(fun=mean,geom="line")+geom_jitter()+ylab("%other_rented+tenures")+theme_dark()
p8/p7
```



```
combined_plot = (p4/p3)|(p6/p5)|(p8/p7) + plot_layout(guides = "collect")
modelsummary::datasummary_skim(PaneLong)
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
year	6	0	1996.0	17.4	1971.0	1996.0	2021.0	
owner_occupied	30	0	1645.6	1039.8	126.0	1877.0	3247.0	
local_authority_etc	29	0	927.5	817.2	8.0	680.5	3353.0	
other_rented	30	0	1770.1	1251.7	144.0	1612.0	4750.0	
other_tenures	24	0	64.7	61.8	0.0	57.0	206.0	
graduate	30	0	2218.2	2318.8	9.0	1325.0	7804.0	
not_graduate	30	0	14 135.9	22 403.0	254.0	6410.0	82 770.0	
gentrified	2	0	0.4	0.5	0.0	0.0	1.0	
location	N	%						
ancoats	6	20.0						
chettam	6	20.0						
crumpsall	6	20.0						
kersal	6	20.0						
quays	6	20.0						